

Finale Messwerte

-O0 & hardware unoptimized (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.056679s

K1 Optimized: 0.027160s

K1 Speedup: 2.09x

--- Start Kernel 2 ---

K2 Baseline: 0.009158s

K2 Optimized: 0.008027s

K2 Speedup: 1.14x

--- Start Kernel 3 ---

K3 Baseline: 0.570567s

K3 Unrolled: 0.556960s

K3 Sorted: 77.452052s

K3 Speedup(A): 1.02x

K3 Speedup(B): 0.01x

--- Start Kernel 4 ---

K4 Baseline: 44.071876s (sum=64222.91)

K4 Optimized: 44.240834s (sum=64222.90)

K4 Speedup: 1.00x

=== MEASUREMENT FINISHED ===

-O2 & hardware unoptimized (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.015929s

K1 Optimized: 0.012834s

K1 Speedup: 1.24x

--- Start Kernel 2 ---

K2 Baseline: 0.007375s

K2 Optimized: 0.007146s

K2 Speedup: 1.03x

--- Start Kernel 3 ---

K3 Baseline: 0.517663s

K3 Unrolled: 0.513900s

K3 Sorted: 24.216098s

K3 Speedup(A): 1.01x

K3 Speedup(B): 0.02x

--- Start Kernel 4 ---

K4 Baseline: 1.092015s (sum=64222.91)

K4 Optimized: 1.271494s (sum=64222.90)

K4 Speedup: 0.86x

=== MEASUREMENT FINISHED ===

-O3 & hardware unoptimized (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.015919s

K1 Optimized: 0.007434s

K1 Speedup: 2.14x

--- Start Kernel 2 ---

K2 Baseline: 0.007166s

K2 Optimized: 0.007126s

K2 Speedup: 1.01x

--- Start Kernel 3 ---

K3 Baseline: 0.517691s

K3 Unrolled: 0.514088s

K3 Sorted: 24.241826s

K3 Speedup(A): 1.01x

K3 Speedup(B): 0.02x

--- Start Kernel 4 ---

K4 Baseline: 1.092115s (sum=64222.91)

K4 Optimized: 1.270319s (sum=64222.90)

K4 Speedup: 0.86x

=== MEASUREMENT FINISHED ===

-O0 & hardware unoptimized (but caches enabled) (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.002562s

K1 Optimized: 0.002682s

K1 Speedup: 0.96x

--- Start Kernel 2 ---

K2 Baseline: 0.000366s

K2 Optimized: 0.000315s

K2 Speedup: 1.16x

--- Start Kernel 3 ---

K3 Baseline: 0.052839s

K3 Unrolled: 0.064846s

K3 Sorted: 4.243774s

K3 Speedup(A): 0.81x

K3 Speedup(B): 0.01x

--- Start Kernel 4 ---

K4 Baseline: 0.109029s (sum=64222.91)

K4 Optimized: 0.117707s (sum=64222.90)

K4 Speedup: 0.93x

=== MEASUREMENT FINISHED ===

-O2 & hardware unoptimized (but caches enabled) (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.002562s

K1 Optimized: 0.002684s

K1 Speedup: 0.95x

--- Start Kernel 2 ---

K2 Baseline: 0.000366s

K2 Optimized: 0.000314s

K2 Speedup: 1.17x

--- Start Kernel 3 ---

K3 Baseline: 0.052833s

K3 Unrolled: 0.064860s

K3 Sorted: 4.243732s

K3 Speedup(A): 0.81x

K3 Speedup(B): 0.01x

--- Start Kernel 4 ---

K4 Baseline: 0.109058s (sum=64222.91)

K4 Optimized: 0.117730s (sum=64222.90)

K4 Speedup: 0.93x

=== MEASUREMENT FINISHED ===

-O3 & hardware unoptimized (but caches enabled) (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.001142s

K1 Optimized: 0.001380s

K1 Speedup: 0.83x

--- Start Kernel 2 ---

K2 Baseline: 0.000269s

K2 Optimized: 0.000265s

K2 Speedup: 1.02x

--- Start Kernel 3 ---

K3 Baseline: 0.050406s

K3 Unrolled: 0.057717s

K3 Sorted: 2.455224s

K3 Speedup(A): 0.87x

K3 Speedup(B): 0.02x

--- Start Kernel 4 ---

K4 Baseline: 0.107380s (sum=64222.91)

K4 Optimized: 0.116081s (sum=64222.90)

K4 Speedup: 0.93x

=== MEASUREMENT FINISHED ===

-O0 & Full HW-Optimization (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.002302s

K1 Optimized: 0.002677s

K1 Speedup: 0.86x

--- Start Kernel 2 ---

K2 Baseline: 0.000092s

K2 Optimized: 0.000122s

K2 Speedup: 0.75x

--- Start Kernel 3 ---

K3 Baseline: 0.003692s

K3 Unrolled: 0.003412s

K3 Sorted: 3.809674s

K3 Speedup(A): 1.08x

K3 Speedup(B): 0.00x

--- Start Kernel 4 ---

K4 Baseline: 0.003044s (sum=64222.91)

K4 Optimized: 0.003133s (sum=64222.90)

K4 Speedup: 0.97x

=== MEASUREMENT FINISHED ===

-O3 & Full HW-Optimization (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.001143s

K1 Optimized: 0.001382s

K1 Speedup: 0.83x

--- Start Kernel 2 ---

K2 Baseline: 0.000030s

K2 Optimized: 0.000025s

K2 Speedup: 1.22x

--- Start Kernel 3 ---

K3 Baseline: 0.001850s

K3 Unrolled: 0.001857s

K3 Sorted: 2.382360s

K3 Speedup(A): 1.00x

K3 Speedup(B): 0.00x

--- Start Kernel 4 ---

K4 Baseline: 0.002573s (sum=64222.91)

K4 Optimized: 0.001757s (sum=64222.90)

K4 Speedup: 1.46x

=== MEASUREMENT FINISHED ===

-O0 & Full HW-Optimization WITHOUT Caches (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.048158s

K1 Optimized: 0.026839s

K1 Speedup: 1.79x

--- Start Kernel 2 ---

K2 Baseline: 0.001824s

K2 Optimized: 0.001074s

K2 Speedup: 1.70x

--- Start Kernel 3 ---

K3 Baseline: 0.055909s

K3 Unrolled: 0.045680s

K3 Sorted: 64.788917s

K3 Speedup(A): 1.22x

K3 Speedup(B): 0.00x

--- Start Kernel 4 ---

K4 Baseline: 42.994568s (sum=64222.91)

K4 Optimized: 42.984214s (sum=64222.90)

K4 Speedup: 1.00x

=== MEASUREMENT FINISHED ===

-O3 & Full HW-Optimization WITHOUT Caches (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.015073s

K1 Optimized: 0.007431s

K1 Speedup: 2.03x

--- Start Kernel 2 ---

K2 Baseline: 0.000248s

K2 Optimized: 0.000229s

K2 Speedup: 1.08x

--- Start Kernel 3 ---

K3 Baseline: 0.017753s

K3 Unrolled: 0.013943s

K3 Sorted: 22.516791s

K3 Speedup(A): 1.27x

K3 Speedup(B): 0.00x

--- Start Kernel 4 ---

K4 Baseline: 0.015392s (sum=64222.91)

K4 Optimized: 0.012178s (sum=64222.90)

K4 Speedup: 1.26x

=== MEASUREMENT FINISHED ===

-O0 & Full HW-Optimization WITHOUT FPU (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.002301s

K1 Optimized: 0.002681s

K1 Speedup: 0.86x

--- Start Kernel 2 ---

K2 Baseline: 0.000094s

K2 Optimized: 0.000122s

K2 Speedup: 0.77x

--- Start Kernel 3 ---

K3 Baseline: 0.016210s

K3 Unrolled: 0.016233s

K3 Sorted: 3.829597s

K3 Speedup(A): 1.00x

K3 Speedup(B): 0.00x

--- Start Kernel 4 ---

K4 Baseline: 0.030907s (sum=64222.91)

K4 Optimized: 0.036659s (sum=64222.90)

K4 Speedup: 0.84x

=== MEASUREMENT FINISHED ===

-O3 & Full HW-Optimization WITHOUT FPU (Array Size = 4096)

=== START OF MEASUREMENT ===

--- Start Kernel 1 ---

K1 Baseline: 0.001142s

K1 Optimized: 0.001380s

K1 Speedup: 0.83x

--- Start Kernel 2 ---

K2 Baseline: 0.000030s

K2 Optimized: 0.000024s

K2 Speedup: 1.25x

--- Start Kernel 3 ---

K3 Baseline: 0.014876s

K3 Unrolled: 0.015068s

K3 Sorted: 2.401218s

K3 Speedup(A): 0.99x

K3 Speedup(B): 0.01x

--- Start Kernel 4 ---

K4 Baseline: 0.028682s (sum=64222.91)

K4 Optimized: 0.033044s (sum=64222.90)

K4 Speedup: 0.87x

=== MEASUREMENT FINISHED ===